UNIT 12 MOBILITY, PORTABILITY, REPLICATION AND CLUSTERING

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12.0 INTRODUCTION

Mobile devices are used for various use cases in our daily lives. We use mobile devices to chat with our friends, to take pictures, to send and receive emails, play games, navigate through maps, to attend online meetings and such. Modern day smart phones provide machine learning (ML) and artificial intelligence (AI) enabled features such as vison, image recognition, augmented reality and many other features. Customers, enterprises, businesses all use mobile apps to deliver information, services and to share resources.

The mobile device accesses the services over the Internet and uses enterprise applications, cloud databases and other services. To provide constant connectivity and data access mobile device has to be in constant touch with the access points.

In this chapter we discuss the mobile data management, data replication and adaptive clustering related to mobile device.

12.1 OBJECTIVES

After going through this unit, you should be able to

- understand key concepts of mobile data management,
- understand data replication schemes,
- understand key concepts of adaptive clustering

12.2 MOBILE DATA MANAGEMENT

Mobile devices manage the data required for its services and apps locally as well on remote secure cloud databases. The mobile data includes the contact list, photos, media files, conversations, documents, game data and such. Mobile device stores these files in local database and also regularly synchronizes the files with the remote database.

A mobile app might use the data that is distributed across various databases. For instance, a gaming mobile app might use the authentication service from the enterprise's active directory and it might use the cloud database to persist the game details (such as game levels, game points for a given user). Hence the mobile device has to process the queries for distributed databases. We have depicted a mobile computing system in Figure 1 that uses distributed services and database across cloud platform and corporate data center and business services.

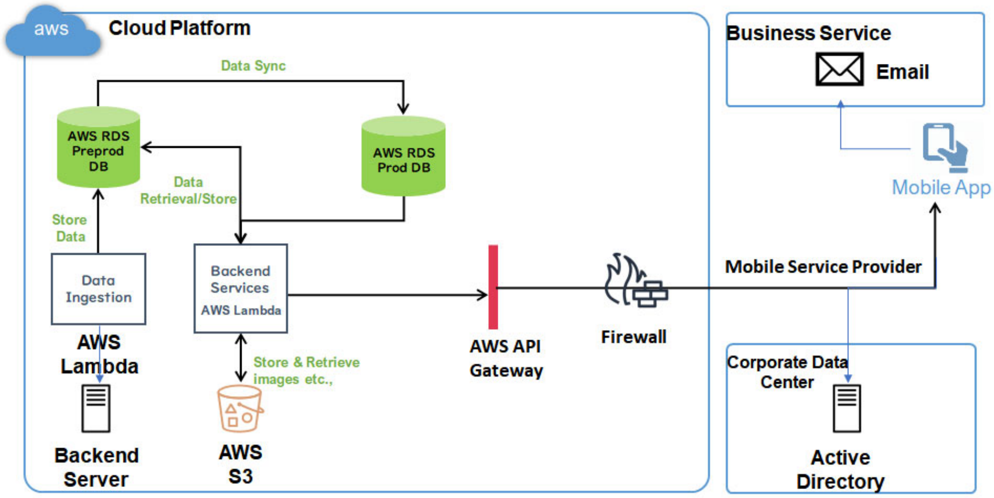


Figure 1 Mobile computing system that uses services distributed across cloud and data centre.

In Figure 1, the mobile app is connected over secured network to the cloud platform, corporate data center and business application that is exposed as software as service platform.

The mobile app provides shopping experience to the end user and connects to AWS (Amazon Web Service) cloud platform for backend processing. To start with, the shopping mobile app authenticates the user against the active directory hosted in the corporate data center. Once the user is successfully authenticated, the product details and user search results are fetched from the backend severs hosted in the AWS. The product list and product search APIs are hosted in the AWS API Gateway that internally uses the Lambda functions and backend servers to get the required information. The backend servers and the Lambda function retrieves the information from the AWS RDS database service. The mobile app users use the email service hosted as software as service platform. In this example, the shopping mobile app has to query against three separate systems and process the information retrieved from these systems.

Mobile app has to manage the data from various sources in its local database. We have given the popular mobile databases below –

- SQLite is one of the most popular mobile databases as it can efficiently store and process different formats of data. SQLite can run in the mobile devices as it uses minimal resources.

- AWS DynamoDB is one of the most popular NoSQL databases that can efficiently store the key-value pair data. Mobile apps use AWS DynamoDB to store the session data, shopping cart data and other key-value data pair.

- MongoDB is a popular opensource database for storing and efficiently managing the documents.

- Redis is a popular caching platform that can manage the frequently used data for use cases such as leaderboards, gaming applications.

- Neo4J is a popular graph database that can efficiently query the relationship between data.

12.2.1 Challenges in Mobile Data Management

The data requirements for Mobile apps are unique as they need to manage the large data volume with low latency and over unpredictable networks. Given below are the main challenges faced by the mobile databases –

Data Synchronization

The data has to be stored be stored in the local database as well on the remote database. We need to design the efficient data synchronization process to synchronize the local data with the remote database on the cloud.

Transaction Management

As the mobile apps work with distributed databases, maintaining the data integrity, ACID (atomicity, consistency, integrity and durability), strict consistency and the correlation of data across all the databases becomes a challenge.

Data processing

The mobile apps need to identify what data that needs to be processed on the client side and what data has to be processed on the server side. The data that needs heavy duty processing has to be done at the remote server end to ensure high performance.

Network Connectivity

Due to the dynamic nature of the mobile network, the mobile apps need to handle the data management in unpredictable network conditions. One of the popular approaches is to use offline-first approach wherein the data is stored locally during low network scenarios and it will be synchronized with the remote database when the network stabilizes.

Data Security

The data at rest and data in transit has to be secured through encryption, transport level security and other mechanisms.

Managing Location Dependency

Mobile apps need to handle the queries that are location dependent and need to manage other dependencies.

Resource management

As the mobile device has limited compute, storage and battery capacity, the mobile app has to manage the data within these constraints.

12.2.2 Handling Mobility Issues

Mobile devices are mostly on the move which brings in its own set of challenges and issues. When the mobile devices are on the move the strength of the signal may vary and sometimes the network may not be available. Another challenge is that the mobile battery may be low and hence the background job and the mobile app should work in low power mode. Few of the mobile apps (such as restaurant finder, hospitals near me) need to be location aware and hence the query processing and data processing has to be location aware.

Mobility also impacts the power consumption of the mobile device. The mobile battery consumption varies based on the signal strength which also has to be factored in.

Location-aware computing during mobility

Given below are some of the main location aware computing that need to be handled as part of the mobility –

- Change of the cell tower due to change in location.
- Change of signal strength and hence change in battery consumption.
- Change in the wireless carrier during change of region (such as during international travel).
- Change of communication protocol (such as 5G to 4G)
- Change of business application (such as change from email to online meeting)
- Change in the service endpoint (such as corporate data center to cloud platform)
- Change of protocol (such as HTTP to UDP)
- Change of network (such as GSM to CDMA)
- Change in local security restrictions (such as restrictions to pre-paid plan in a given region)
- Change in time zone and its impact on time-based apps (such as calendar)
- Change in cost due to change of network.

Challenges and issues during mobility

Given below are the main issues and challenges due to mobility –

- Challenges in varying bandwidth and unpredictable network and the impact of network-intensive apps (such as maps, car rental apps)
- Varying power consumption due to the signal strength impacts the background apps and the resource intensive apps
- Handling transaction and the related data across various service endpoints and across heterogenous databases pose challenges.
- Handling the data backup jobs (such as photo sync up) during change in location.
- On-going call or video conference might get impacted due to change in the signal strength.
- Data synchronization between local database and remote database need to be handled when the network changes.

12.2.3 Handling Wireless Communication

Mobile device communicates with the nearest node of the cell for data communication. Based on the device location, the network may be partially or fully unavailable (such as at remote areas or in escalator). Additionally the available bandwidth is unpredictable which is based on the location and the amount of traffic at the given location.

Given below are the main factors of wireless communication that impacts mobility –

- The network bandwidth and the signal strength based on the location.
- Availability of network connectivity based on the location.
- Changing network topology (such as 3G to Wi-Fi)
- Limited battery power and compute and storage on the local mobile device.

12.2.4 Handling Mobile Portability

As mobile devices are resource constrained, the mobile devices need to manage the large amount of data generated by the mobile apps that require large storage and heavy-duty data processing.

Data portability involves replicating or moving the data across different servers, environments and across different operating systems.

Given below are the core best practices for handling mobile portability –

- Store the frequently used data in local data store (such as in SQLIte database) and synchronize the data from local data store to cloud database on frequent basis.
- Encrypt the data during transit to ensure data security.
- Backup the local data to remote servers on regular basis (such as for photos, videos and media files)
- Centrally store the core data (such as contact list, browser bookmarks, chat conversations) for security reasons.
- Use an open-standard data format to provide flexible and extensible data replication (such as using JSON or XML file formats).
- Cache the frequently used data for optimal performance.
- Adopt offline-first approach to manage data locally when there are no networks.

☞ Check Your Progress 1

1. ________ database is used for managing documents.
2. _______ database is used for efficiently querying the relationships.
3. _______ is a popular caching framework that can be used for managing the frequently used values.
4. Local database and remote database is kept up to date through ___________.

12.3 DATA REPLICATION SCHEMES

Mobile device stores the data of many mobile apps. A camera app stores the captured photos and videos whereas a messaging app stores the shared files in the mobile device. As the size of the stored files increase, we need to manage the risk of data loss, security and durability.

Data replication is a mechanism wherein the data on the mobile device is replicated to another computer or device or remote cloud server. If the copied data changes in the target (such as remote cloud server), the updated data is replicated back to the mobile device. In summary the changed data is synchronized across all the devices.

Given below are the main advantages of data replication -

- Data replication ensure the high durability of data
- Data replication improves the reliability and availability of the data.
- Data replication optimizes the latency (as data is stored nearer to the location of access) thereby improving the data access performance.
- Data replication allows the queries to use the data from a single location thereby reducing the query complexity.

12.3.1 Data Replication Methods

There are mainly three kinds of data replication methods. In the one-to-one data replication method as depicted in Figure 2, the user uploads the data to a server or PC and it gets replicated to the mobile device. If the data changes either in the server or in the mobile device, the updated data is synchronized to the other system. An example of one-to-one data replication is the user specific data backup (such as files, media files) between user's PC and user's mobile device.

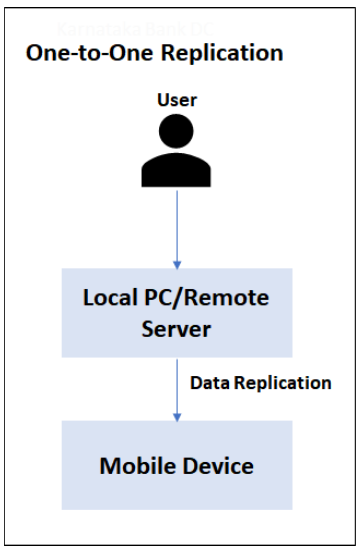


Figure 2 One-to-one replication

In a one-to-many replication scheme, the user copies the data to the centralized server which then replicates the data to various mobile devices as depicted in Figure 3. An example of one-to-many data replication is the security policy replication between an organizational server and all the manged mobile devices belonging to its employees.

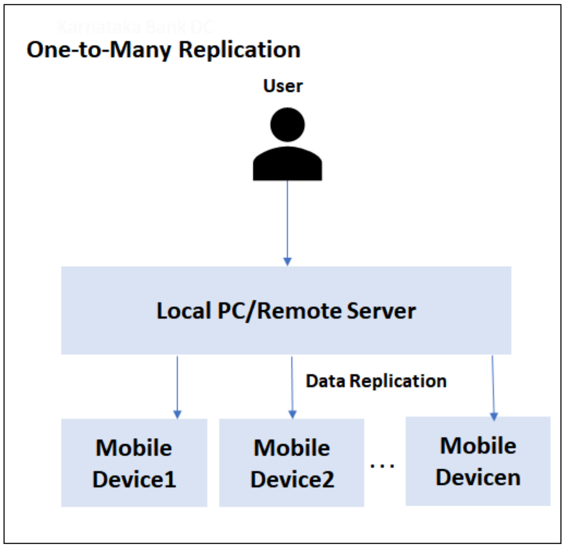
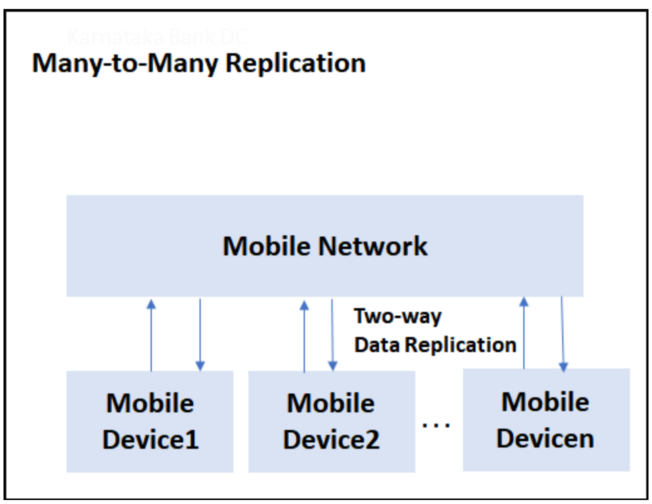


Figure 3 One-to-many replication

In many to many data replication, multiple mobile devices perform bi-directional replicational of the data through the mobile network as depicted in Figure 4. Multi-player game is an example of many-to-many data replication wherein the game data is continuously replicated across all the mobile devices of the participating players.



12.3.1 Considerations for Data Replication

As data replication involves copy data across various devices, we need to factor in various design considerations for data replication. We have given the key design considerations for data replication below –

- Data synchronization task - We need to design the data synchronization job that handles the incremental data replication across devices. The data synchronizer task has to synchronize the data at specified time intervals.

- Data merging and conflict resolution – During many to many data synchronization as the data from multiple device needs to be synchronized, the synchronizer has to handle the conflict resolution (when the same object has been updated simultaneously by two different devices) and data merging (merging the changes).

- Data security – The data replication job should also ensure that data is securely copied to the target locations.

- Handling data schema changes – Different domains have different data schemas and as such the data replication job has to ensure that a common schema based on open standards (such as JSON or XML) is used for data replication.

- Handling data conversion – Due to the heterogenous nature of the devices, the data may need to be converted during the data replication process.

12.3.2 Data Replication Types

There are broadly two main types of replication – full replication and partial replication.

In full replication the entire data is replicated across devices. For instance, the mobile device copies all the images from the device to the cloud drive. In the partial replication, only the limited set of data to the target device. For instance, the mobile device replicates only the data updated on the day to the central server on daily basis.

Pull and push synchronization

In traditional client-server kind of scenario, the server pushes the data on a frequent basis. The client device caches the pushed data in its local data store and uses the data for its mobile apps. The client mobile device regularly pulls the updated data from the server. An example for this model is a gaming mobile app wherein the game mobile app running on the local device gets the complete set of game rules during the one-time setup. Post the mobile game app installation, the app regularly pulls the updated game rules from the centralized rules database on daily basis. This method is also known as “one-to-many” communication wherein a single server pushes the data and many clients pull the data.

The second model of synchronization is that a single device will pull and push the information as in the case of peer-to-peer communication. An example for this model is a peer-to-peer sharing mobile app that shares the data and files to another mobile device.

Synchronous and Asynchronous replication

In synchronous replication, the data is replicated continuously in real time synchronously. Synchronous replication is used in strongly consistent use cases. For instance, a stock trading application replicates the transaction data in real time.

Asynchronous replication replicates the data asynchronously in batch mode. This improves the performance and optimizes the

12.4 ADAPTIVE CLUSTERING

Adaptive clustering is a dynamic grouping of mobile devices to improve performance and optimize the energy consumption. The grouping of mobile devices happen based on their location proximity and communication protocols.

In order to conserve the consumed energy, the mobile devices communicate with its nearest neighbors instead of transmitting the signal to all the mobile devices. By localizing the communication, we can optimize the latency and conserve the energy.

12.4.1 Adaptive Clustering Process

In this section we have explained the process of adaptive clustering

Cluster formation

Various factors such as the mobile device's signal strength, distance, communication protocols, network load are used along with clustering algorithms (such as distance based clustering, traffic load based clustering, mobility based clustering) to create the clusters.

Device communication

Post cluster creation, the mobile devices in a given cluster communicate directly. To communicate with other clusters the mobile devices use cluster heads that act as gateway to across the clusters.

Cluster reconfiguration

The cluster is re-adjusted dynamically based on the changing conditions. For instance, if the mobile device moves out of the location range, the mobile device is removed from the cluster and added to a new cluster to ensure optimal performance and optimal energy consumption.

We have depicted a sample adaptive clustering in Figure 4

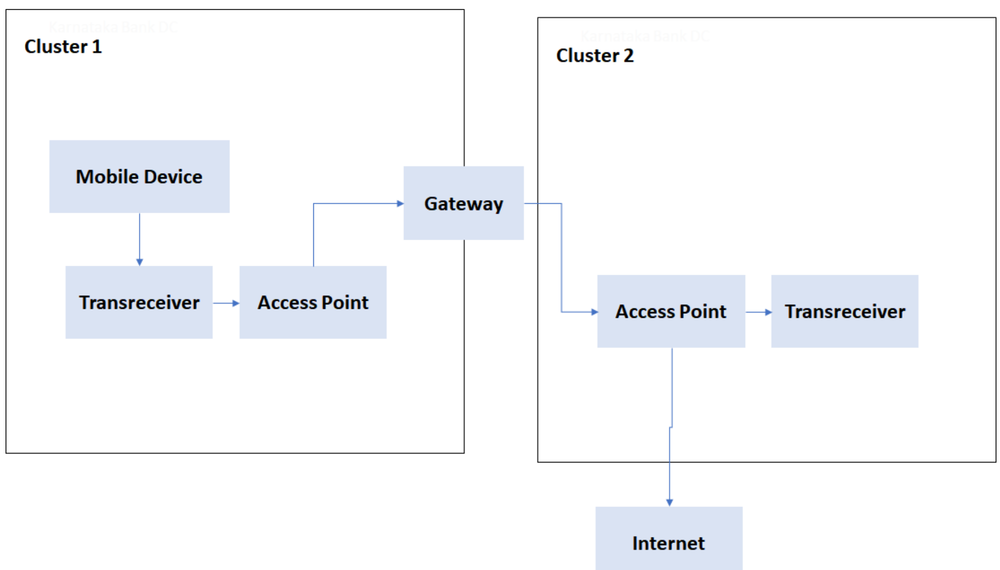


Figure 4 Sample Adaptive clustering

☞ Check Your Progress 2

1. In _____ replication method, the data from the server or PC and gets replicated to the mobile device.
2. In ______replication method, the data from a centralized server replicates the data to various mobile devices.
3. In _____ replication method bi-directional replication happens across various devices.
4. In __________replication, the data is replicated continuously in real time.
5. _______ clustering is a dynamic grouping of mobile devices to improve performance and optimize the energy consumption.

12.5 SUMMARY

In this chapter we discussed various concepts related to data handling schemes of mobile devices. Mobile devices use data distributed across various datastores. Mobile app uses various database such as SQLite (for managing data locally), MongoDB (for document management), Redis (for caching), DynamoDB (as NoSQL database for key-value pair), Neo4J (for graph use cases). The main challenges in mobile data management are Data synchronization , Transaction management ,Data processing, Network connectivity, Data security, Managing location dependency and Resource management. Varying bandwidth, varying power consumption, handling transactions, handling data backup jobs are the key Challenges and issues during mobility. The core best practices for handling mobile portability are local data caching, data encryption, data backup, open standards, caching. The three kinds of data replication methods are one-one, one-to-many and many-to-many. Adaptive clustering is a dynamic grouping of mobile devices to improve performance and optimize the energy consumption.

12.6 SOLUTIONS/ANSWERS

Check Your Progress 1

1. mongodb.
2. Neo4j.
3. Redis .
4. Data synchronization

Check Your Progress 2

1. One-to-one.
2. One-to-many
3. Many-to-many
4. synchronous
5. adaptive

12.7 FURTHER READINGS

References

https://en.wikipedia.org/wiki/Network_virtualization

https://en.wikipedia.org/wiki/Cloud_computing